

# Generation Interconnection Request System Impact Study Report

## For Queue Project Q18-0936

Tac-Distributed Energy Resource Int 138 kV  
March 28, 2019

### 1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q18-0936. The Interconnection Customer proposes a 35.8 MW battery energy storage facility interconnecting at the Tac-Distributed Energy Resource Int 138 kV substation in the New Jersey area. The request was received on February 02, 2018. The study evaluated system intact and single contingency conditions using the regional planning case, including 387 higher-queued active requests and their assigned network upgrades.

### 2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 5 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$561,000.

### 3. Network Impacts

#### 3.1 Generator Deliverability and Single Contingency Analysis

| Monitored Facility                                            | Contingency                                            | Loading (%) | Rating (MVA) | DFAX (%) |
|---------------------------------------------------------------|--------------------------------------------------------|-------------|--------------|----------|
| MONTOUR STATION SUBSTATION<br>500-345 kV Autotransformer      | CORONET-WESTWOOD GENERATION<br>LLC SUBSTATION 500.0 kV | 141.7       | 883          | 6.8      |
| JH WARDEN 500/345kV Transformer                               | CORONET-WESTWOOD GENERATION<br>LLC 500.0 kV            | 106.9       | 678          | 5.3      |
| TALENERGY MARTINS CREEK LLC<br>HARWOOD-FELIDA 345.0 kV        | FELIDA SUBSTATION 500/345kV Auto                       | 102.5       | 1,479        | 8.6      |
| TALENERGY MARTINS CRK LLC<br>HARWOOD SUBSTATION 345-1 kV Auto | FELIDA 500/345 kV Auto                                 | 102.5       | 1,482        | 7.3      |
| TALENERGY MARTINS CRK LLC<br>HARWOOD 500/1kV Auto             | FELIDA 500/345kV Transformer                           | 101.4       | 1,497        | 7.3      |

#### 3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

#### 3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

### 4. Required Network Upgrades and Cost Estimates

| Facility                                                 | Required Upgrade                              | Cost Estimate | Allocation to Project |
|----------------------------------------------------------|-----------------------------------------------|---------------|-----------------------|
| MONTOUR STATION SUBSTATION<br>500-345 kV Autotransformer | Replace transformer with<br>higher-rated unit | \$35,818,000  | 0% (\$151,000)        |

| Facility                                                     | Required Upgrade                           | Cost Estimate | Allocation to Project |
|--------------------------------------------------------------|--------------------------------------------|---------------|-----------------------|
| JH WARDEN 500/345kV Transformer                              | Replace transformer with higher-rated unit | \$21,706,000  | 1% (\$144,000)        |
| FELIDA-TALENENERGY MARTINS CRK LLC HARWOOD 345.0 kV          | Reconductor line                           | \$9,350,000   | 0% (\$47,000)         |
| TALENENERGY MARTINS CRK LLC HARWOOD 345/1kV Transformer Bank | Replace transformer with higher-rated unit | \$19,741,000  | 0% (\$88,000)         |
| TALENENERGY MARTINS CRK LLC HARWOOD 500/1kV Auto             | Replace transformer with higher-rated unit | \$29,347,000  | 0% (\$131,000)        |
|                                                              | Total allocated to project                 |               | \$561,000             |

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.